

# Modern Algorithmic Game Theory

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# Recap

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Last week, we started talking about simultaneous decision making. We covered:

- normal-form games
- general- and zero-sum games
- pure and mixed strategies
- expected utilities and best responses
- iterated removal of dominated strategies

Today, we will introduce **solution concepts**, namely **Maximin strategies** and **Nash equilibria**.

# Solution Concepts

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(a) Maximin



(b) Nash Equilibrium



**Maximin Strategy**

# Maximin

## Definition: Maximin Strategy

A Maximin strategy of player  $i$  is a strategy that guarantees the highest possible expected utility against the worst-case opponent. We define it as:

$$\arg \max_{\pi_i \in \Pi_i} \min_{\pi_{-i} \in \Pi_{-i}} u_i(\pi_i, \pi_{-i}) = \arg \max_{\pi_i \in \Pi_i} BRV_i(\pi_i)$$

- We assume everyone else is *“out there to get us”*
- A maximin policy maximizes our expected utility assuming the worst-case scenario
- We also use  $v_i = \max_{\pi_i} \min_{\pi_{-i}} u_i(\pi_i, \pi_{-i})$  to denote the Maximin value of player  $i$

# Maximin in Pure Strategies

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|           | cooperate  | defect     |
|-----------|------------|------------|
| cooperate | $(-1, -1)$ | $(-3, 0)$  |
| defect    | $(0, -3)$  | $(-2, -2)$ |

Table: Prisoner's Dilemma

Let's reason from the row player's perspective:

- the worst-case payoff when playing cooperate is -3 and when playing defect is -2
- therefore, the strategy maximizing the worst-case payoff is defect

# Maximin in Pure Strategies

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|          | rock | paper | scissors |
|----------|------|-------|----------|
| rock     | 0    | -1    | 1        |
| paper    | 1    | 0     | -1       |
| scissors | -1   | 1     | 0        |

Table: Rock Paper Scissors

Let's reason from the row player's perspective:

- the worst-case payoff is always -1, no matter what action we choose
- however, we can do better than this when we consider mixed strategies
- the uniform strategy gives the expected utility of 0, but how can we find it?

# Optimizing Against Best Response

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- In two-player zero-sum games, it holds that

$$\arg \max_{\pi_i \in \Pi_i} \min_{\pi_{-i} \in \Pi_{-i}} u_i(\pi_i, \pi_{-i}) = \arg \max_{\pi_i \in \Pi_i} u_i(\pi_i, b(\pi_i))$$

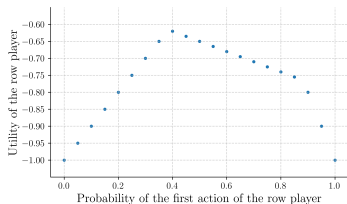
- We are optimizing against a best-responding opponent
- Let's consider the following  $2 \times 3$  zero-sum game

|     | A  | B  | C    |
|-----|----|----|------|
| X   | -1 | 0  | -0.8 |
| 1-X | 1  | -1 | -0.5 |

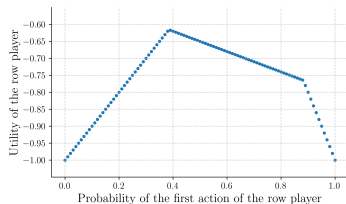
- The worst-case payoff for the row player is again -1 no matter what action they play
- Let's visualize the best response value function  $f(\pi_i) = u_i(\pi_i, b(\pi_i))$

# Best Response Value Function

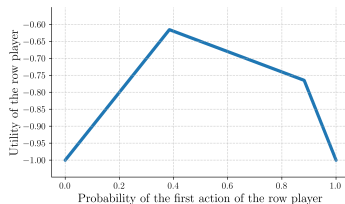
- We can see that placing the probability of around 0.385 on the first action yields a much better expected utility of around -0.6!
- The piece-wise linear structure of the best response value function can be leveraged to create a linear program that finds the Maximin strategy (and its value) that maximizes the function



(a) Step size 0.05



(b) Step size 0.01



(c) Step size 0.001



# **Nash Equilibrium**

# Nash Equilibrium

## Definition: Nash Equilibrium

Strategy profile  $(\pi_i, \pi_{-i})$  is a Nash equilibrium if none of the players can benefit from unilaterally deviating from their policy. Mathematically,

$$\forall i \in \mathcal{N}, \forall \pi'_i \in \Pi_i : u_i(\pi_i, \pi_{-i}) \geq u_i(\pi'_i, \pi_{-i})$$

- The central solution concept in game theory with extremely diverse applications
- Represents a stable solution where players cannot individually improve their utility
- We can also look at this as each player playing a best response against other players
- It is easy to verify – all strategies must be best responses
- It has been proven that every game with a finite number of players, each having a finite number of actions, always has a mixed-strategy Nash equilibrium

# Nash Equilibrium in Pure Strategies

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|           | cooperate  | defect     |
|-----------|------------|------------|
| cooperate | $(-1, -1)$ | $(-3, 0)$  |
| defect    | $(0, -3)$  | $(-2, -2)$ |

Table: Prisoner's Dilemma

- Only a single Nash equilibrium, both players play defect, even though playing cooperate would lead to a higher expected utility for both players

|      | stop      | go           |
|------|-----------|--------------|
| stop | $(0, 0)$  | $(-1, 1)$    |
| go   | $(1, -1)$ | $(-10, -10)$ |

Table: The Game of Chicken

- Two pure Nash equilibria – (stop, go) and (go, stop)

# Nash Equilibrium in Pure Strategies

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|          | rock | paper | scissors |
|----------|------|-------|----------|
| rock     | 0    | -1    | 1        |
| paper    | 1    | 0     | -1       |
| scissors | -1   | 1     | 0        |

Table: Rock Paper Scissors

- We can naively try finding a pure strategy Nash equilibrium by enumerating all pure strategy profiles and verifying each strategy is a best response
- However, pure strategy Nash equilibrium might not exist!
- Consider Rock Paper Scissors, for any pure action of one player, their opponent has profitable deviation  $\rightarrow$  no pure strategy Nash equilibrium

# Support Enumeration

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We now present a brute-force algorithm that can find all Nash equilibria, both pure and mixed, in any nondegenerate two-player normal-form game by enumerating and checking all pairs of supports.

- Recall the **best response condition** that states that all actions in the support of a best response strategy are also best responses and so have the same expected utilities
- Therefore, finding NEs essentially boils down to finding the right pairs of supports
- For a given pair of supports, we can compute the probabilities of individual actions in the supports by solving a system of linear equations<sup>1</sup>
- After solving the system of equations, we still need to check that actions outside of the supports do not lead to higher expected utilities; if they did, the given pair of supports and the found strategies would not form a NE in the original game

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<sup>1</sup>For games with three or more players, the system of equations becomes non-linear

# Support Enumeration

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- For some support and a strategy vector  $\pi_i$  (where actions outside of the support have probability 0) consider elements of the following payoff vector  $\pi_i A_{-i}$
- The elements of the payoff vector  $\pi_i A_{-i}$  corresponding to actions in the opponent's support must all equal the best-response value  $v_{-i} = \max_{\pi_{-i}} u_{-i}(\pi_i, \pi_{-i})$  of the **opponent**, while the remaining elements must not exceed it
- In other words, the player needs to mix actions in their support so that the values of actions in the opponent's support are best-responding (and thus all the same)
- Now, consider the following game with supports  $\{X, Y\}$  and  $\{A, B\}$

|   | A          | B          | C          |
|---|------------|------------|------------|
| X | (0, 0)     | (0, 1)     | (-10, -10) |
| Y | (1, 0)     | (-10, -10) | (-10, -10) |
| Z | (-10, -10) | (-10, -10) | (-10, -10) |

# Support Enumeration

|   | A          | B          | C          |
|---|------------|------------|------------|
| X | (0, 0)     | (0, 1)     | (-10, -10) |
| Y | (1, 0)     | (-10, -10) | (-10, -10) |
| Z | (-10, -10) | (-10, -10) | (-10, -10) |

- The row player needs to mix actions in their support  $\{X, Y\}$  so that the values of the column player for actions in their support  $\{A, B\}$  are all the same

$$0\pi_1(X) + 1\pi_1(Y) = v_2$$

$$1\pi_1(X) - 10\pi_1(Y) = v_2$$

- By the same reasoning for the column player, we get

$$0\pi_2(A) + 0\pi_2(B) = v_1$$

$$1\pi_2(A) - 10\pi_2(B) = v_1$$

- Solution:  $\pi_1(X) = 0.909091 = \pi_2(A)$  and  $\pi_1(Y) = 0.090909 = \pi_2(B)$

# Support Enumeration

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|   | A          | B          | C          |
|---|------------|------------|------------|
| X | (0, 0)     | (0, 1)     | (-10, -10) |
| Y | (1, 0)     | (-10, -10) | (-10, -10) |
| Z | (-10, -10) | (-10, -10) | (-10, -10) |

- Repeating the same steps as on the previous slides but this time for supports  $\{X, Z\}$  and  $\{A, B\}$ , we get:

$$0\pi_1(X) - 10\pi_1(Z) = v_2$$

$$1\pi_1(X) - 10\pi_1(Z) = v_2$$

- ... and for the column player:

$$0\pi_2(A) + 0\pi_2(B) = v_1$$

$$-10\pi_2(A) - 10\pi_2(B) = v_1$$

- There's no solution and therefore no NE candidate for the given supports

An aerial photograph of a wide, frozen river. The river is mostly covered in a light blue-grey ice, with darker, more textured areas of water or thinner ice interspersed. A large, irregularly shaped island or peninsula is located in the upper center of the frame. The river flows from the top towards the bottom, with several bends and channels. A semi-transparent, light yellow rectangular box is centered over the middle of the image, containing the text "General-Sum Games" in a dark blue, sans-serif font.

## General-Sum Games

# Maximin Strategies vs Nash Equilibria

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- Maximin is defined as a strategy, whereas Nash equilibrium as a strategy profile
- Are the two solution concepts the same?
- Why would we want to play them?

# Non-Rational Opponents, Deviating from NEs

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Let's elaborate on the properties of Nash equilibria.

- Suppose we play against a non-rational opponent.
- A non-rational player does not necessarily try to maximize their utility, they can play arbitrary strategies
- Consider a Nash equilibrium  $(\pi_1, \pi_2)$  and say we decide to play  $\pi_1$ , what are our guarantees?
- Even though  $\pi_2$  maximizes our opponent's utility, they can make mistakes and select different (non-equilibristic) strategy  $\pi'_2$
- Choosing a different strategy than  $\pi_2$  cannot lead to a better expected utility for the opponent
- However, it can lead to a much worse outcome for us, it can be the case that  $u_1(\pi_1, \pi_2) \gg u_1(\pi_1, \pi'_2)$  and so opponent's mistakes can hurt us

# Deviating from Nash Equilibria

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- Consider Prisoner's Dilemma and assume we play our equilibrium strategy (defect)
- A rational opponent would also play defect leading to our reward of -2, but a non-rational opponent can also choose cooperate which leads to our reward of 0
- Now consider The Game of Chicken and a NE strategy profile (go, stop)
- Say, we play go but our non-rational opponent makes a mistake and also plays go, leading to our reward of -10 instead of 1

|           | cooperate  | defect     |
|-----------|------------|------------|
| cooperate | $(-1, -1)$ | $(-3, 0)$  |
| defect    | $(0, -3)$  | $(-2, -2)$ |

Table: Prisoner's Dilemma

|      | stop      | go           |
|------|-----------|--------------|
| stop | $(0, 0)$  | $(-1, 1)$    |
| go   | $(1, -1)$ | $(-10, -10)$ |

Table: The Game of Chicken

# Rational Players, Multiple Equilibria Problem

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- Suppose there are two optimal strategy profiles in the game  $(\pi_1, \pi_2^a)$  and  $(\pi_1, \pi_2^b)$
- The opponent is indifferent between their two strategies, they do not care which strategy they choose (given our strategy  $\pi_1$ ), since  $u_2(\pi_1, \pi_2^a) = u_2(\pi_1, \pi_2^b)$
- However, we might care! We can imagine a situation where  $u_1(\pi_1, \pi_2^a) \neq u_1(\pi_1, \pi_2^b)$
- Even though both players play optimally, different optimal strategies can lead to different utilities!
- Consider the following game with two pure NEs - (go, plan a) and (go, plan b)
- In either of the two pure equilibria, the column player's expected utility is the same; however, the row player's utility is considerably different

|      | plan a  | plan b |
|------|---------|--------|
| go   | (10, 5) | (1, 5) |
| wait | (0, 0)  | (0, 0) |

An aerial photograph of a frozen river or lake. The ice is white and textured, with dark, winding channels of water or meltwater visible. A semi-transparent yellow rectangular box is centered in the lower half of the image, containing the text "Zero-Sum Games" in a dark blue, sans-serif font.

## Zero-Sum Games

# Maximin and Minimax Values

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We will now investigate the important relationship between the maximin values  $v_i$  and  $v_{-i}$  of players  $i$  and  $-i$ , respectively. We already know that

$$v_i = \max_{\pi_i} \min_{\pi_{-i}} u_i(\pi_i, \pi_{-i})$$

$$v_{-i} = \max_{\pi_{-i}} \min_{\pi_i} u_{-i}(\pi_i, \pi_{-i})$$

However, we can also show that

$$\begin{aligned} v_{-i} &= \max_{\pi_{-i}} \min_{\pi_i} -u_i(\pi_i, \pi_{-i}) \\ &= \max_{\pi_{-i}} - \max_{\pi_i} u_i(\pi_i, \pi_{-i}) \\ &= - \min_{\pi_{-i}} \max_{\pi_i} u_i(\pi_i, \pi_{-i}) \end{aligned}$$

# Minimax Theorem

## Theorem

For any two-player zero-sum game, the following holds

$$\max_{\pi_i} \min_{\pi_{-i}} u_i(\pi_i, \pi_{-i}) = \min_{\pi_{-i}} \max_{\pi_i} u_i(\pi_i, \pi_{-i})$$

- The Minimax theorem states a critical result – the maximin and the minimax values are in balance  $v_i = -v_{-i}$ .
- We refer to this unique value as the **game value** and denote it by  $GV_i$
- The theorem was proven by John von Neumann in 1928 and has dramatic consequences for two-player zero-sum games
- Von Neumann himself later wrote *“As far as I can see, there could be no theory of games ... without that theorem ... I thought there was nothing worth publishing until the Minimax theorem was proved.”*

# Nash Equilibrium $\Leftrightarrow$ Maximin Strategies

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**Theorem: Nash equilibrium  $\implies$  Maximin strategies**

For any two-player zero-sum game and for any pair of mixed strategies  $\pi_1, \pi_2$ , it holds that if  $(\pi_1, \pi_2)$  forms a Nash equilibrium then  $\pi_1$  and  $\pi_2$  are both Maximin strategies.

**Theorem: Maximin strategies  $\implies$  Nash equilibrium**

For any two-player zero-sum game and for any pair of mixed strategies  $\pi_1, \pi_2$ , it holds that if  $\pi_1$  and  $\pi_2$  are both Maximin strategies then  $(\pi_1, \pi_2)$  forms a Nash equilibrium.

# Key Motivating Property

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## Theorem

If we follow an optimal policy when playing from both positions, the expected utility against any opponent is greater than or equal to zero:

$$(\pi_i, \pi_{-i}) \in Nash : u_i(\pi_i, \pi'_{-i}) + u_{-i}(\pi'_i, \pi_{-i}) \geq 0 \forall \pi'_i, \pi'_{-i}$$

# Summary of the Solutions Concepts

The following table summarizes the existence guarantees, and the complexity of verifying and computing the two solution concepts in general- and zero-sum two-player games.

We consider Maximin strategy to be a strategy that maximizes a player's worst-case payoff. However, in zero-sum games we consider it to be a strategy that attains the unique game value  $GV_i$ , which in general requires mixed Maximin strategies.

|              | General-Sum |      | Zero-Sum |      |
|--------------|-------------|------|----------|------|
|              | Maximin     | Nash | Maximin  | Nash |
| Pure         | ✓           | ×    | ×        | ×    |
| Mixed        | ✓           | ✓    | ✓        |      |
| Verification | Easy        | Easy | Easy     |      |
| Computation  | Easy        | Hard | Easy     |      |

# Week 2 Homework

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You can find more detailed descriptions of homework tasks in the GitHub repository.

1. Best response value function
2. Support enumeration
3. Prove that Nash equilibrium  $\Leftrightarrow$  Maximin strategies in zero-sum games